

Achieve crypto-agility and build quantum resilience

Protect sensitive data, mitigate risk and prepare for quantum resilience through crypto-agility.



Highlights

Enabling crypto-agility with unified cryptography management

Visibility through discovery and inventory across hybrid cloud

Risk assessment, compliance, and policy management

Centralized lifecycle management and remediation

Protection of assets and cryptographic objects

The digital landscape is evolving faster than organizations can secure, leaving critical data exposed. Every transaction, workload, and device relies on cryptography, a way of protecting information by turning it into a code that only authorized people or systems can read. Yet fragmented tools and manual processes make it hard to know where it's used, if it's secure, or how to update it.

Quantum computing, artificial intelligence (AI), and hybrid cloud adoption are reshaping data security and exposing new layers of cryptographic risk. Quantum computing will fundamentally transform data protection, forcing enterprises to rethink how they secure systems and information.

Yet only 30% of organizations have completed a cryptographic inventory, and just 19% have set near-term maturity goals for quantum-safe readiness, despite 73% reporting collaboration between business and technology leaders on this issue.¹ Meanwhile, AI is creating a new, often unprotected, attack surface, adding an average USD 670,000 in breach costs from shadow-AI-related security incidents.² The clock is ticking as quantum-capable systems are expected to break current encryption in the early 2030s, when the first vulnerable algorithms will be deprecated by the National Institute of Standards and Technology (NIST).³

This creates an urgent need for organizations to act now to achieve crypto-agility and quantum resilience—the ability to rapidly adapt cryptographic mechanisms in response to threats, technological advances, or vulnerabilities without disrupting infrastructure or business processes.

30%

of organizations have completed a cryptographic inventory.

19%

Of organizations have set near-term maturity goals for quantum-safe readiness.

Why crypto-agility

Achieving crypto-agility and quantum resilience starts with removing the obstacles that slow transformation. Today's technology landscape exposes weaknesses in how organizations manage, secure, and update cryptography.

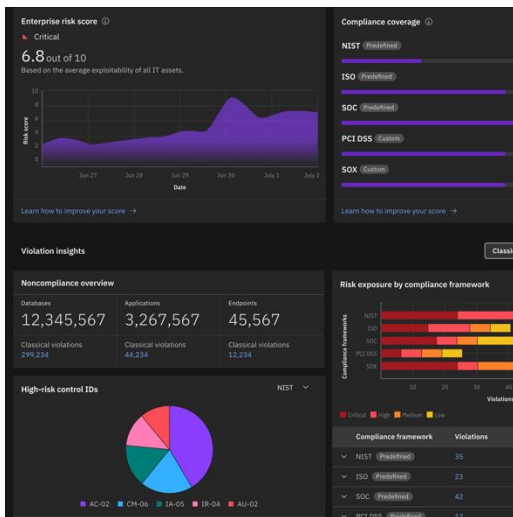
Organizations face growing pressure from upcoming quantum-safe deadlines set by NIST, the UK National Cyber Security Centre (NCSC), and other international bodies. At the same time, AI-related data exposure, data sprawl and poor visibility across fragmented systems increase risk. TLS certificate validity periods have now been shortened to just 47 days, creating a continuous cycle of renewals that many organizations struggle to manage efficiently.

Cryptography constantly becomes outdated. Algorithms weaken, regulations evolve, and new vulnerabilities—quantum or otherwise—emerge continuously. Yet updating cryptography remains slow, costly, and risky: every algorithm or key change can require rewriting code, retesting integrations, and redeploying services across an entire organization.

To stay ahead, enterprises need a crypto-agile foundation that abstracts cryptography from applications and enables secure, organization-wide updates to happen centrally without disruption. By introducing an enterprise-wide cryptography management platform, cryptographic functions become centralized, standardized, and easily updated.

Teams can rotate cryptographic objects or replace algorithms from a single control point and push updates across applications, services, and processes automatically. This “update once, protect everywhere” approach eliminates manual effort, reduces operational risk, and delivers long-term ROI through faster adaptation to evolving threats.

While quantum computing presents the next systemic challenge, the real advantage lies in being able to adapt quickly to any future cryptographic exposure.



Enabling crypto-agility with unified cryptography management

IBM Guardium® Cryptography Manager helps your organization achieve crypto-agility through discovery and inventory, risk assessment, cryptographic object lifecycle management, compliance readiness, and remediation of cryptographic objects. This enables you to manage cryptography at scale, reduce risk, and move toward quantum-safe resilience—all from one solution. With Guardium Cryptography Manager you unlock:

Visibility through discovery and inventory across hybrid cloud

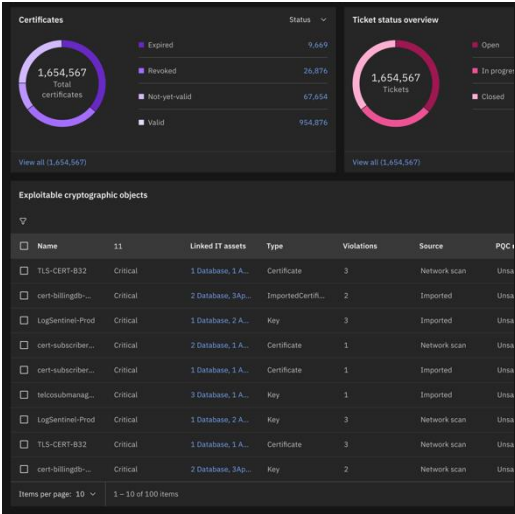
Continuously discover and inventory IT assets and the cryptographic objects that secure them—keys, certificates, secrets, algorithms, and SSH public keys—across on-premises and cloud environments. Automatically catalog keys and certificates, scan application source code for quantum-vulnerable cryptography, and visualize risk posture through a consolidated dashboard. Centralize cryptographic visibility across distributed environments to reduce data sprawl driven by rapid cloud and AI adoption. Integrations with Nmap, Kubernetes, Qualys, Jira, and ServiceNow support automated response workflows. Discovery and inventory also extend into code-centric cryptographic risk through import profiles for IBM Quantum Safe Explorer scan findings. Teams can ingest findings for code repositories and code assets, use Portfolio View to monitor imported repositories, analytics findings, and CBOM data in one place, and expand coverage across more systems through an open plug-in framework and custom transformers for non-standard file formats.

Risk assessment, compliance, and policy management

Continuously detect weak, expired, or misconfigured assets through posture monitoring and policy enforcement. Prioritize vulnerabilities and accelerate remediation with custom policies that include compliance tags and control IDs, helping teams map violations to internal and industry frameworks. Assess post-quantum cryptography readiness, support migration to NIST-approved quantum-safe algorithms, and maintain alignment with global standards such as HIPAA, PCI DSS, and GDPR while generating audit-ready compliance reporting. Policy workflows also provide richer violation context, including filtering, side-panel details, and exploitability explanations to support faster remediation decisions.

Centralized certificate lifecycle management and remediation

Simplify complexity through unified key and certificate lifecycle management covering issuance, retrieval, rotation, renewal, and revocation. Automate renewals at both policy and certificate lifecycle management level, deploy certificates discovered through Kubernetes, and create CA-signed certificates for Venafi PKI to reduce outage risk and manual effort. Dynamic groups and fine-grained access control further help teams organize assets, apply policies efficiently, and support secure remediation at scale.



Centralized transparent database encryption key management

Centralize transparent database encryption key management to safeguard data at rest across major database environments. Guardium Cryptography Manager simplifies operations with centralized key lifecycle control, strengthens compliance, and includes Microsoft SQL Server TDE support for broader database coverage.

Protection of assets and cryptographic objects

Extend crypto-agility from posture to protection with controls that help safeguard applications, endpoints, and traffic as cryptographic standards evolve. Adaptive Proxy protects applications and endpoints from “harvest-now, decrypt-later” attacks without code modifications or architectural overhaul. Forward Proxy secures outbound traffic and helps teams understand how cryptographic algorithms affect behavior and performance. Together, these controls help maintain compliance, protect sensitive data, and preserve confidentiality.

From insight to action with AI and automation

Accelerate remediation and operations with AI-driven and agent-based automation. Guardium Cryptography Manager integrates with IBM watsonx.ai for AI-assisted ticket content and supports MCP Server for natural language queries and agent-based workflows, helping teams retrieve asset counts, identify certificates nearing expiration, and review active policy violations with less manual effort.

Together, these controls help maintain compliance and preserve data confidentiality as cryptographic standards evolve.

Quantum resilience is no longer a future concern. It's a present requirement.

IBM Guardium Cryptography Manager helps enterprises achieve both crypto-agility and quantum resilience, uniting discovery, policy, lifecycle management, and protection under one modern, scalable solution. It empowers teams to manage cryptography across hybrid cloud environments, strengthen data protection, and accelerate readiness for quantum-resilient operations.

For more information

To learn more about our solution, contact your IBM representative or IBM Business Partner, or visit ibm.com/products/guardium-cryptography-manager.

1. IBV Report: Secure the post-quantum future, IBM, 2025
2. Cost of a Data Breach 2025, IBM, 2025
3. Transition to PQC Standards, NIST, 2024

© Copyright IBM Corporation 2026

Produced in the
United States of America
March 2026

IBM, the IBM logo, and Guardium, are trademarks or registered trademarks of International Business Machines Corporation, in the United States and/or other countries. Other product and service names might be trademarks of IBM or other companies. A current list of IBM trademarks is available on ibm.com/trademark.

This document is current as of the initial date of publication and may be changed by IBM at any time. Not all offerings are available in every country in which IBM operates.

THE INFORMATION IN THIS DOCUMENT IS PROVIDED "AS IS" WITHOUT ANY WARRANTY, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

IBM products are warranted according to the terms and conditions of the agreements under which they are provided.

